Area ratios without lengths

L2-X1 Level 2 Intermediate · Geometry · Beyond the textbook

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Areas as unknowns and ratios on lines as equations: same-height triangles, cevians, trapezia and parallelograms, and the triangle formed inside by three cevians.
By the end you can turn a ratio on a line into a ratio of areas, and back, using a common apex or a common base; find unknown areas in a triangle cut by cevians by labelling them and solving linear equations; use the area ratios of a trapezium cut by its diagonals, and find the triangle formed inside by three cevians.
Prerequisites: Area, Pythagoras and similar shapes (L2-GE2). **Contests:** IMC, Cayley, Hamilton, Intermediate Kangaroo, AMC 10. **Time:** about 75 minutes.

1 Lesson

Triangle ABC has area 30. The point D lies on BC with $BD = 2DC$, and E is the midpoint of AD. Find the area of triangle BEC. Triangles ABD and ADC have the same height from A, so their areas are in the ratio $BD : DC = 2 : 1$, giving $[ABD] = 20$ and $[ADC] = 10$. Since E is the midpoint of AD, the same argument with apex B gives $[BED] = \frac{1}{2}[ABD] = 10$, and with apex C gives $[CED] = 5$. Hence $[BEC] = 15$. No length in the problem is known, and none was needed.

Here $[XYZ]$ denotes the area of triangle XYZ. The fact used above is Proposition 1 of Book VI of Euclid's *Elements*: triangles under the same height are to one another as their bases. School courses use area as $\frac{1}{2} \times$ base $\times$ height with numbers for the lengths, while contest problems often give only ratios, or only areas, and ask for another ratio or area. The *area method* treats areas as the unknowns and ratios on lines as the equations. It is a standard topic in Chinese olympiad training, and Zhang Jingzhong developed it into a complete method that was later used for computer proofs (Chou, Gao and Zhang, *Machine Proofs in Geometry*, 1994). We assume L2-GE2: areas by decomposition and similar triangles.

1.1 Four facts

Fact 1 (same height).
If X, Y, Z, W lie on one line and P does not, then $[PXY] : [PZW] = XY : ZW$. In particular, a cevian AD of triangle ABC gives $[ABD] : [ADC] = BD : DC$.
Fact 2 (cevian ratio). If P is a point of the cevian AD other than A, then $[ABP] : [ACP] = BD : DC$.
Fact 3 (common base). If the segments AD and XY meet at P, then $[AXY] : [DXY] = AP : PD$.
Fact 4 (common angle). If D lies on AB and E on AC, then $\frac{[ADE]}{[ABC]} = \frac{AD}{AB} \cdot \frac{AE}{AC}$.

Proofs. Fact 1: both triangles have the distance from P to the line as height. Fact 2: let $BD : DC = r$. By Fact 1 with apex A, $[ABD] = r[ACD]$, and with apex P, $[PBD] = r[PCD]$. Subtracting, $[ABP] = [ABD] - [PBD] = r([ACD] - [PCD]) = r[ACP]$. Fact 3: drop perpendiculars AH and

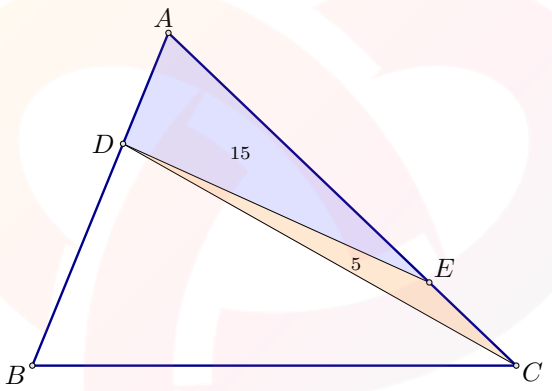
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DK to line XY. The right-angled triangles AHP and DKP have equal angles at P, so they are similar and $AH : DK = AP : PD$; these are the heights of AXY and DXY on the common base XY. Fact 4: by Fact 1, $[ADE] = \frac{AD}{AB}[ABE]$ and $[ABE] = \frac{AE}{AC}[ABC]$. ■

Fact 2 is the one most students do not know, and it is the one that sets up the equations in Section 3: a single ratio on a line is shared by two pairs of triangles, one pair with apex A and one with apex P. Fact 3 gives the ratio in which a cevian is cut by another line, which is what is needed for the area of the triangle formed inside by three cevians (Example 4).

Example 1.
Triangle ABC has area 60. The point D lies on AB with $AD : DB = 1 : 2$, and E lies on AC with $AE : EC = 3 : 1$. Find the areas of triangles ADE and DEC.

Solution.
By Fact 4, $[ADE] = \frac{1}{3} \cdot \frac{3}{4} \cdot 60 = 15$. For DEC we use apex D and the line AC: by Fact 1, $[DEC] : [DAE] = EC : AE = 1 : 3$, so $[DEC] = 5$. As a check, $[ADC] = \frac{1}{3} \cdot 60 = 20 = 15 + 5$.



1.2 Trapezia and parallelograms

Fact 5 (trapezium).
Let $ABCD$ be a trapezium with $AB \parallel DC$ and $AB = k \cdot DC$, whose diagonals meet at X. Then

$$[ABX] : [BCX] : [CDX] : [DAX] = k^2 : k : 1 : k.$$

In any convex quadrilateral with diagonals meeting at X, $[ABX] \cdot [CDX] = [BCX] \cdot [DAX]$.

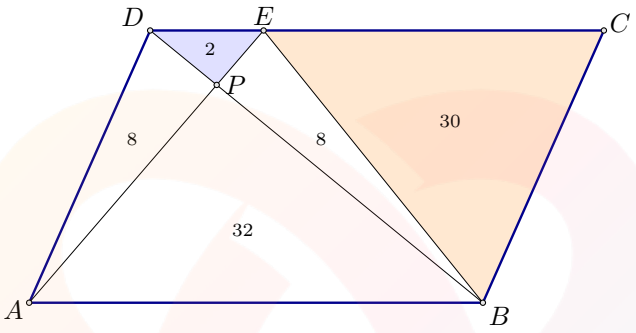
Proof. For any convex quadrilateral, Fact 1 with apex B and then apex D gives $[ABX] : [BCX] = AX : XC = [DAX] : [CDX]$, which is the second statement. In the trapezium, triangles ABX and CDX are similar (alternate angles), with ratio $AB : CD = k$, so $AX : XC = k$ and $[ABX] = k^2[CDX]$. Also $[ABD] = [ABC]$, since they share the base AB and D, C are at the same distance from it; removing $[ABX]$ from both gives $[DAX] = [BCX]$. Finally $[BCX] = \frac{XC}{AX}[ABX] = \frac{1}{k} \cdot k^2[CDX] = k[CDX]$. ■

A parallelogram is cut by a diagonal into two halves, and any triangle with one side of the parallelogram as base and its third vertex on the opposite side has half the area. A segment from a vertex to a point of a non-adjacent side creates a trapezium, and Fact 5 applies to it.

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Example 2.
Parallelogram $ABCD$ has area 80. The point E lies on CD with $DE : EC = 1 : 3$, and AE meets the diagonal BD at P. Find the area of triangle DPE and of quadrilateral $PBCE$.

Solution.
Since $AB \parallel DE$, the quadrilateral $ABED$ is a trapezium whose diagonals AE and BD meet at P, and $AB = 4\,DE$. By Fact 5 the triangles ABP, BEP, EDP, DAP have areas in the ratio $16 : 4 : 1 : 4$, so $[ABED]$ is 25 parts. Triangle BCE has base $EC = \frac{3}{4}DC$ and the height of the parallelogram, so $[BCE] = \frac{3}{4} \cdot 40 = 30$ and $[ABED] = 50$. One part is 2. Hence $[DPE] = 2$ and $[PBCE] = [BPE] + [BCE] = 8 + 30 = 38$.
The similar triangles DPE and BPA give $DP : PB = 1 : 4$ directly; the trapezium gives all four pieces at once.



1.3 Labelling the unknown areas

When two cevians BE and CF meet at P, the triangle is cut into four pieces, and drawing AP makes five triangles. The method is as follows.

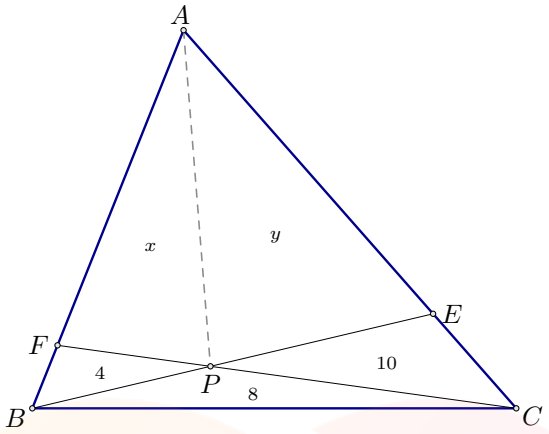
1. Draw the segment from the remaining vertex to the intersection point, so that every region is a triangle.
2. Write a letter for each unknown area.
3. For each cevian through P, read the ratio in which P divides it from a pair of known triangles (Fact 1), and write the same ratio for the pair of triangles with apex at the third vertex (Fact 2 applied to that vertex). Each cevian gives one equation.
4. Solve the linear equations.

Example 3.
The cevians BE and CF of triangle ABC meet at P. The areas of triangles BPF, BPC and CPE are 4, 8 and 10. Find the area of quadrilateral $AFPE$.

Solution.
No length is known, and a coordinate solution would have to choose a shape for the triangle with six unknown coordinates. Draw AP and let $x = [APF]$, $y = [APE]$. Two unknowns, so we need two equations, one from each cevian.
On CF: triangles BPF and BPC have apex B, so $FP : PC = 4 : 8 = 1 : 2$. Triangles APF and APC have apex A and the same bases, so $x : (y + 10) = 1 : 2$. On BE: triangles BPC and CPE have apex C, so $BP : PE = 8 : 10 = 4 : 5$, and with apex A, $(x + 4) : y = 4 : 5$.

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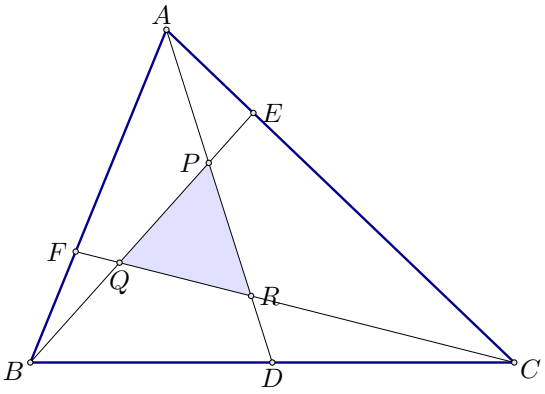
The equations are $2x = y + 10$ and $5x + 20 = 4y$. Substituting $y = 2x - 10$ gives $5x + 20 = 8x - 40$, so $x = 20$, $y = 30$ and $[AFPE] = 50$. The whole triangle has area 72, and you can check that $AF : FB = [AFC] : [BFC] = 60 : 12 = 5 : 1$.



Example 4.
In triangle ABC of area 30, D is the midpoint of BC, E lies on CA with $CE : EA = 3 : 1$, and F lies on AB with $AF : FB = 2 : 1$. Let AD meet BE at P, BE meet CF at Q, and CF meet AD at R. Find the area of triangle PQR.

Solution.
First the standard way. Area ratios do not depend on the shape of the triangle (a shear keeps every area, and a stretch in one direction multiplies every area by the same factor), so we could take $A = (0,0)$, $B = (3,0)$, $C = (0,20)$, find P, Q, R by solving three pairs of linear equations, and finish with the shoelace formula. This works and takes a page of fractions.
With areas, note that the triangles ABP, BCQ, CAR and PQR fill triangle ABC without overlapping, so $[PQR] = 30 - [ABP] - [BCQ] - [CAR]$. Each of the three corner triangles needs one ratio along a cevian, and Fact 3 gives it.
For P, use the common base BE: $AP : PD = [ABE] : [DBE]$. Now $[ABE] = \frac{1}{4} \cdot 30 = \frac{15}{2}$ and $[CBE] = \frac{45}{2}$, so $[DBE] = \frac{1}{2}[CBE] = \frac{45}{4}$. Hence $AP : PD = 2 : 3$ and $[ABP] = \frac{2}{5}[ABD] = \frac{2}{5} \cdot 15 = 6$.
For Q, use the base CF: $[BCF] = \frac{1}{3} \cdot 30 = 10$, $[ACF] = 20$ and $[ECF] = \frac{3}{4} \cdot 20 = 15$, so $BQ : QE = 10 : 15 = 2 : 3$ and $[BCQ] = \frac{2}{5}[BCE] = \frac{2}{5} \cdot \frac{45}{2} = 9$.
For R, use the base AD: $[CAD] = 15$ and $[FAD] = \frac{2}{3}[BAD] = 10$, so $CR : RF = 3 : 2$ and $[CAR] = \frac{3}{5}[CAF] = \frac{3}{5} \cdot 20 = 12$.
Therefore $[PQR] = 30 - 6 - 9 - 12 = 3$, one tenth of the triangle. The same three steps give the inner triangle for any three cevians; the general answer is Routh's theorem, which is not worth memorising for IMC or AMC 10.

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Common traps.

- Using Fact 1 for triangles whose bases are not on one line, or whose apex is not shared. Before writing a ratio, name the apex and the line.
- Writing $FP : PC$ as a ratio of the triangles on either side of P without a common apex, for example $[BPF] : [CPE]$.
- In a trapezium, the triangles on the parallel sides are in the ratio k^2, and the two on the legs are equal. It is easy to use k instead of k^2.
- Assuming that three cevians cut a triangle into pieces of equal area because the figure looks symmetric.
- Forgetting to draw the segment to the third vertex in Section 3: the quadrilateral $AFPE$ is not a triangle, and Fact 2 needs triangles.

1.4 How to spot this in a contest

1. The question gives only ratios on sides, or only areas of some regions, and asks for another area or ratio.
2. A figure with cevians meeting inside a triangle and numbers written in the regions.
3. A trapezium or parallelogram with its diagonals, or a segment from a vertex to the opposite side.
4. The answer is a fraction of the whole area: you may then fix the total area as any convenient number, such as the least common multiple of the denominators.
5. A coordinate solution needs you to choose a shape. If the answer cannot depend on the shape, look for an area argument first.

2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

Problem 1 $\star$ Triangle ABC has area 84. The point D lies on BC with $BD : DC = 3 : 4$, and E lies on AD with $AE : ED = 2 : 1$. Find the area of triangle ABE.

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Problem 2 ⋆⋆ $ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 2DC$. The point M is the midpoint of AB, and DM meets the diagonal AC at P. What fraction of the area of $ABCD$ is the area of triangle APD?

Problem 3 ⋆⋆ The cevians AD and BE of triangle ABC meet at P. The areas of triangles ABP, BPD and APE are 12, 8 and 6. Find the area of quadrilateral $PDCE$.

Problem 4 ⋆⋆⋆ Triangle ABC has area 70. The point D lies on BC with $BD : DC = 1 : 3$, E lies on CA with $CE : EA = 2 : 3$, and F is the midpoint of AB. The segments AD, BE, CF bound a small triangle. Find its area.

Problem 5 ⋆⋆⋆ Parallelogram $ABCD$ has area 140. The point E lies on AB with $AE : EB = 1 : 2$, and F is the midpoint of BC. The segments DE and AF meet at P. Find the area of triangle APD.

Problem 6 ⋆⋆⋆ $ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 3DC$. Its diagonals meet at X, and the line through X parallel to AB meets AD at M and BC at N. Find $\frac{[ABNM]}{[ABCD]}$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **Purple Comet 2010 Middle School Problem 14** ⋆
   Two of the four triangles at P are given: what is k, and what are the other two?
2. **PMWC 2000 Problem T6** ⋆⋆
   Work outwards from FGH: one application of Fact 1 for each ratio given, ending with apex A on line BC.
3. **AMC 10A 2009 Problem 23** ⋆⋆
   What do equal areas of AED and BEC say about the sides AB and CD?
4. **AMC 8 2019 Problem 24** ⋆⋆
   Label the small triangles and use ratios along BD and along BC.
5. **NYSML Spring 1977 Problem 1** ⋆⋆
   Fact 3 with a common base gives where each segment cuts the median.
6. **AMC 10A 2016 Problem 19** ⋆⋆
   Each of AE and AF forms a trapezium-type pair of similar triangles with the diagonal BD.
7. **AMC 10B 2016 Problem 19** ⋆⋆⋆
   Find where AC and AG cut EF from pairs of similar triangles with parallel sides.
8. **AIME 1984 Problem 3** ⋆⋆⋆
   The three corner triangles are similar to ABC; compare the square roots of the areas.
9. **UKMT Senior Kangaroo 2024 Question 19** ⋆⋆⋆
   The diagonal QT is parallel to RS and UP; use the trapezium ratios.

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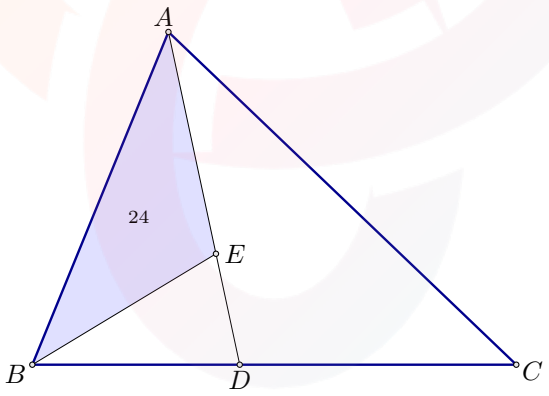
4 Hints

1. Two applications of Fact 1, first with apex A, then with apex B.
2. Compare AM with DC. What kind of quadrilateral is $AMCD$?
3. Draw CP and call the two new areas x and y. Each of the cevians AD and BE gives one equation.
4. Remove three triangles, each with one side of ABC as a side. For each one, Fact 3 gives the ratio along a cevian.
5. Extend AF to meet line DC. You then have a trapezium whose diagonals meet at P.
6. Use Fact 5 to give the four triangles at X areas in units. Then find where M divides AD.

5 Solutions

Problem 1

Triangle ABC has area 84. The point D lies on BC with $BD : DC = 3 : 4$, and E lies on AD with $AE : ED = 2 : 1$. Find the area of triangle ABE.



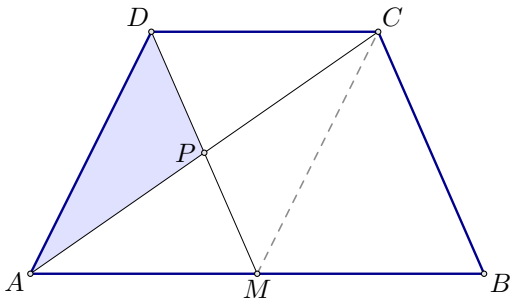
Triangles ABD and ABC have apex A and bases on BC, so $[ABD] = \frac{3}{7} \cdot 84 = 36$. Triangles ABE and ABD have apex B and bases AE and AD on one line, so $[ABE] = \frac{2}{3} \cdot 36 = 24$.

Answer: 24

Problem 2

$ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 2DC$. The point M is the midpoint of AB, and DM meets the diagonal AC at P. What fraction of the area of $ABCD$ is the area of triangle APD?

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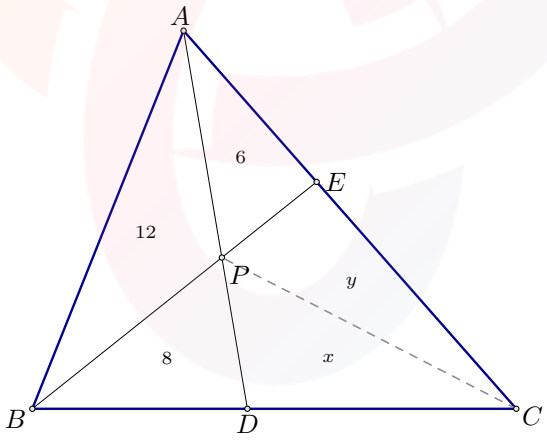


Let $T = [ACD]$. Triangles ABC and ACD have bases AB and DC on parallel lines and the same height, so $[ABC] = 2T$ and $[ABCD] = 3T$. Since $AM = DC$ and $AM \parallel DC$, the quadrilateral $AMCD$ is a parallelogram, and its diagonals AC and DM bisect each other at P. Its area is $[ACD] + [AMC] = T + \frac{1}{2}[ABC] = 2T$, and the diagonals cut it into four triangles of equal area, so $[APD] = \frac{T}{2}$. The fraction is $\frac{T/2}{3T} = \frac{1}{6}$.

Answer: $\frac{1}{6}$

Problem 3

The cevians AD and BE of triangle ABC meet at P. The areas of triangles ABP, BPD and APE are 12, 8 and 6. Find the area of quadrilateral $PDCE$.



Draw CP and let $x = [PDC]$, $y = [PCE]$. Triangles ABP and BPD have apex B, so $AP : PD = 12 : 8 = 3 : 2$; with apex C the same ratio gives $(6 + y) : x = 3 : 2$. Triangles ABP and APE have apex A, so $BP : PE = 12 : 6 = 2 : 1$; with apex C, $(8 + x) : y = 2 : 1$.

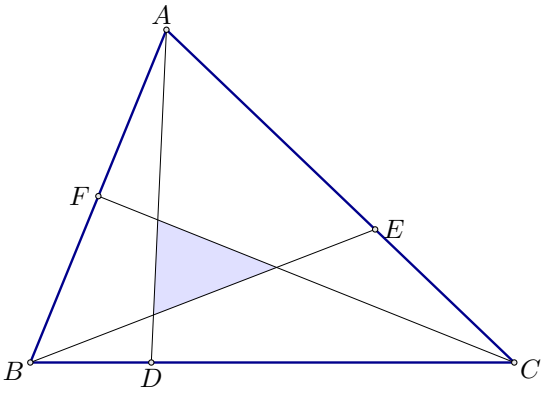
From the second equation $y = \frac{8+x}{2}$, and then the first gives $12 + (8 + x) = 3x$, so $x = 10$ and $y = 9$. The quadrilateral has area 19.

Answer: 19

Problem 4

Triangle ABC has area 70. The point D lies on BC with $BD : DC = 1 : 3$, E lies on CA with $CE : EA = 2 : 3$, and F is the midpoint of AB. The segments AD, BE, CF bound a small triangle. Find its area.

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Let AD meet BE at P, BE meet CF at Q, and CF meet AD at R. The small triangle is PQR, and $[PQR] = 70 - [ABP] - [BCQ] - [CAR]$.

By Fact 1, $[ABE] = \frac{3}{5} \cdot 70 = 42$, $[CBE] = 28$ and $[DBE] = \frac{1}{4} \cdot 28 = 7$. By Fact 3 with base BE, $AP : PD = 42 : 7 = 6 : 1$, so $[ABP] = \frac{6}{7}[ABD] = \frac{6}{7} \cdot \frac{70}{4} = 15$.

Next $[BCF] = [ACF] = 35$ and $[ECF] = \frac{2}{5} \cdot 35 = 14$, so $BQ : QE = 35 : 14 = 5 : 2$ and $[BCQ] = \frac{5}{7}[BCE] = \frac{5}{7} \cdot 28 = 20$.

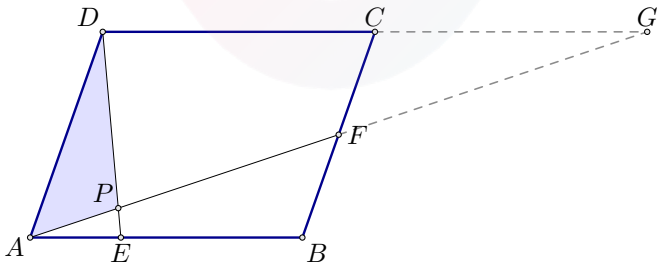
Finally $[CAD] = \frac{3}{4} \cdot 70 = \frac{105}{2}$ and $[FAD] = \frac{1}{2}[BAD] = \frac{35}{4}$, so $CR : RF = 6 : 1$ and $[CAR] = \frac{6}{7}[CAF] = \frac{6}{7} \cdot 35 = 30$.

Hence $[PQR] = 70 - 15 - 20 - 30 = 5$.

Answer: 5

Problem 5

Parallelogram $ABCD$ has area 140. The point E lies on AB with $AE : EB = 1 : 2$, and F is the midpoint of BC. The segments DE and AF meet at P. Find the area of triangle APD.



Extend AF to meet line DC at G. Triangles ABF and GCF have $BF = FC$, equal vertical angles at F and equal alternate angles at B and C, so they are congruent and $CG = AB$. Hence $DG = 2AB$.

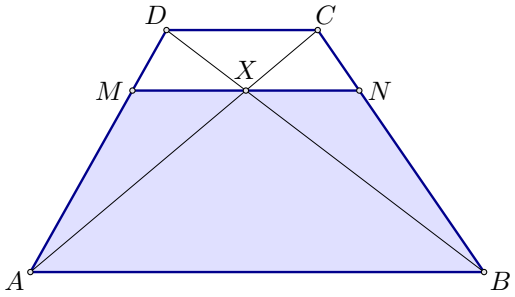
Now $AE \parallel DG$, so $AEGD$ is a trapezium whose diagonals AG and DE meet at P, and $DG : AE = 2AB : \frac{1}{3}AB = 6$. The similar triangles PDG and PEA give $DP : PE = 6 : 1$. Since $[AED] = \frac{1}{3}[ABD] = \frac{1}{3} \cdot 70 = \frac{70}{3}$ (apex D, bases on AB), Fact 1 with apex A gives $[APD] = \frac{6}{7} \cdot \frac{70}{3} = 20$.

Answer: 20

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Problem 6

$ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 3DC$. Its diagonals meet at X, and the line through X parallel to AB meets AD at M and BC at N. Find $\frac{[ABNM]}{[ABCD]}$.



By Fact 5 with $k = 3$, we may take $[CDX] = 1$, $[ABX] = 9$ and $[BCX] = [DAX] = 3$, so $[ABCD] = 16$. In triangle DAB the line MX is parallel to AB, so $DM : DA = DX : DB$. As $DX : XB = DC : AB = 1 : 3$, we get $DM : DA = 1 : 4$. Triangles DMX and DAX have apex X and bases on line DA, so $[DMX] = \frac{1}{4} \cdot 3 = \frac{3}{4}$. In the same way $CN : CB = 1 : 4$ and $[CNX] = \frac{3}{4}$.

The part of the trapezium above MN is made of DMX, CDX and CNX, with total area $\frac{3}{4} + 1 + \frac{3}{4} = \frac{5}{2}$. Hence $[ABNM] = 16 - \frac{5}{2} = \frac{27}{2}$, and the ratio is $\frac{27}{32}$.

Answer: $\frac{27}{32}$